Coding Workflow for RoO Restricted-Input Pairs

General coding rules for every preferential trade agreement (PTA). The definition of an input–output link is expansive: every pair **starts at DV = 1**, and each step turns to 0 only the cases it clearly excludes. Pairs are written **input** → **output** (A → B). The rules include the scope decisions of 19 October 2026; they are summarised in the last section. Results for individual PTAs are reported separately.

Variables

Variable	Contents
pair_id, pta, hs_vintage	Identifiers
input_hs6, output_hs6, descriptions	The pair
DV	Final code; initialised to 1 for every pair
DV_strict	DV × (1 − uncertain): counts only certain 1s
DV_excl_remelt	Robustness version of DV with direct remelting (route = recycle_direct) set to 0
locked	1 if step 2 protects the pair at DV = 1
step	Step that turned the pair to 0 (3–6), or 2 if locked; blank if the pair stayed 1
route	Reason a pair stays 1: same_good, forward, ingredient, component, reconstitute, mixture, recycle_waste, recycle_direct, feed, sibling_intermediate; none if turned to 0
uncertain, uncertain_reason	1 if the pair is kept at 1 but is doubtful: the two AI coders disagree, or the pair falls under a rule that is 1 with uncertainty (step 6); the reason is recorded
source	How the code was reached: rule, rule_textile, ai coders, exact_copy, ml, conflict
dv_ai1, dv_ai2, why_ai1, why_ai2	Each AI coder's code and one-line reason
ml_prob	Model probability (step 7 only)
concord_flag	1 if a code is not one-to-one with the common HS vintage (step 1)

Step 1. Take in the RoO pairs and set DV = 1

- Read each agreement's product-specific rules and list every (restricted input, output) pair at HS6, in the agreement's own HS vintage.
- Map every pair to a common HS vintage with the WCO/UN correlation tables. A code that splits produces one pair per new code; flag non-one-to-one mappings as *concord_flag* = 1 for later checking.
- Remove duplicate pairs within each agreement and keep a pair-level master list across agreements (used in step 7).
- **Set DV = 1 for every pair.** Steps 3–6 can only turn a 1 into a 0; no step turns a 0 back into a 1.

Step 2. Lock identical or same-good pairs at 1

Rule: set *locked* = 1 (*route* = same_good) when input and output are the same good or very similar: the same HS6 code, or the same species, material or variety differing only in processing stage, preservation state, packing or size of presentation, or a treatment the producer can add (roasting, decaffeinating, dyeing, printing, plying). Locked pairs skip steps 3–6.

Input → Output	DV	Why
090111 coffee, not roasted → 090111 coffee, not roasted	1, locked	Same HS6 code (self-pair)
090111 coffee, not roasted → 090121 coffee, roasted	1, locked	Same coffee, roasted
090111 coffee, not roasted → 090112 coffee, decaffeinated	1, locked	Same coffee, decaffeinated
090210 green tea, packs ≤ 3 kg → 090220 green tea, packs > 3 kg	1, locked	Same tea, different packing
520811 unbleached cotton plain weave → 520831 dyed cotton plain weave	1, locked	Same fabric, dyed

Boundary with siblings (step 4). "Very similar" means one can be turned into the other. Goods that differ in a fixed attribute — fibre, species, gender cut, base material — are siblings, not the same good: women's cotton jackets → women's synthetic jackets is not locked and goes to 0 at step 4. The direction must also be possible: dyed fabric → unbleached fabric is not locked and goes to 0 at step 5.

Step 3. Turn to 0: clearly irrelevant pairs

Rule: turn to 0 only when no route from step 6 could apply: the input would have to change into a different species, plant, fibre or substance, and it is not a constituent, feed, part or waste of the output. If in doubt, leave the pair at 1 for step 6.

Input → Output	DV	Why
200941 pineapple juice → 200410 frozen prepared potatoes	1→0	Different plant; juice is not an ingredient of potatoes
210500 ice cream → 210320 tomato ketchup	1→0	Unrelated food preparations
151110 crude palm oil → 151491 crude rapeseed oil	1→0	Refining cannot change the oil's source
710231 unworked diamonds → 711510 platinum catalysts	1→0	A stone does not become a metal
730810 steel bridges → 730729 stainless tube fittings	1→0	Finished articles with unrelated uses
030562 salted cod → 030344 frozen bigeye tuna	1→0	Different fish species, and salted cod is not feed
030379 other frozen fish → 030332 frozen plaice	1→0	Different species; plaice is wild-caught, so no feed route

Caution. Do not turn a pair to 0 at this step when the output is a mixture, a farmed animal or a waste product; leave it at 1 for step 6. Example: chicken cuts → 160249 (prepared pork and mixtures) looks irrelevant but is a mixture pair.

Fibres. Different synthetic fibres — nylon, polyester, acrylic, polypropylene — count as **one** fibre: nylon yarn → acrylic yarn is not a step-3 pair; it is an intermediate sibling (step 4) and stays 1. Different natural or artificial fibres (cotton → polyester, flax → jute, viscose → polyester) remain different fibres.

Fish and other seafood. Different species are different goods: the pair is 0 at this step unless the input could feed the output (step 6, feed). The intermediate-sibling rule of step 4 does not apply to different species.

Step 4. Turn to 0: downstream sibling products

Rule: both goods are **finished** products at the same stage, usually in the same heading or chapter, differing in a fixed attribute (fibre, species, cut, material), and neither physically goes into the other.

Input → Output	DV	Why
620432 women's cotton jackets → 620433 women's synthetic jackets	1→0	Parallel finished garments
620630 women's cotton blouses → 620640 women's man-made-fibre blouses	1→0	Parallel finished garments
640192 waterproof boots → 640110 waterproof boots with metal toe-cap	1→0	Parallel footwear
630622 synthetic-fibre tents → 630621 cotton tents	1→0	Parallel made-up articles
960820 felt-tip pens → 960810 ballpoint pens	1→0	Parallel pens

Exceptions that stay 1 (*route* = component), because one sibling is a part of the other:

Input → Output	DV	Why
640699 footwear parts → 640399 leather footwear	stays 1	Parts (uppers, soles) go into footwear
960891 nibs → 960839 fountain pens	stays 1	Nib is a pen component
960860 ballpoint refills → 960810 ballpoint pens	stays 1	Refill is a pen component
611780 knitted clothing accessories (trims, bows) → 610610 women's knitted cotton blouses	stays 1	Trims are attached to garments

Intermediate siblings stay 1. Step 4 applies only when both goods are finished. Two different intermediate goods of the same material at the same stage stay 1 with *route* = sibling_intermediate. Intermediate goods are fibre, yarn (including sewing thread), fabric, steel and other metal semi-finished products, paper and paperboard in rolls or sheets, glass in the mass and semi-finished glass, feed residues, basic chemicals and parts. Fabrics of the same fibre but a different construction (woven, knitted, pile, nonwoven) are intermediate siblings.

Input → Output	DV	Why
720851 hot-rolled steel sheet → 721420 hot-rolled steel bars	stays 1	Intermediate steel products
480100 newsprint → 480411 kraft paper	stays 1	Intermediate paper grades
480591 other uncoated paper → 480512 straw fluting paper	stays 1	Intermediate paper grades
230230 wheat bran → 230400 soya-bean oil-cake	stays 1	Parallel feed residues
731815 screws and bolts → 731816 nuts	stays 1	Steel fastener parts
520811 cotton woven fabric → 600622 cotton knitted fabric	stays 1	Same fibre, different construction

540241 nylon filament yarn → 540243 polyester filament yarn	stays 1	Synthetic fibres count as one fibre
540110 synthetic sewing thread → 550959 synthetic staple yarn	stays 1	Thread and yarn of one fibre

Step 5. Turn to 0: inverse (backward) pairs

Rule: the input is downstream of the output in the same production chain, so the real relationship runs the other way.

Input → Output	DV	Why
721123 cold-rolled steel → 720229 ferro-silicon	1→0	Ferro-silicon goes into steel, not the reverse
520811 cotton woven fabric → 520511 cotton yarn	1→0	Yarn is woven into fabric
410791 finished leather → 410120 raw bovine hides	1→0	Hides are tanned into leather

- Exception that stays 1: a reverse link through a waste or scrap code (step 6, recycling).
- Exception that stays 1: direct remelting with no scrap code in between (step 6, *route* = recycle_direct).
- Steps 2, 4 and 5 can largely be applied by rule (self-pairs and same-heading stage rules for step 2; sibling and backward rules for steps 4 and 5); AI coders only need to check the exceptions.

Step 6. Turn to 0 only where the expansive definition finds no route

Rule: every pair still at 1 is checked against the expansive routes below. It stays 1 if any route plausibly applies (record the route). It is turned to 0 only if you are sure no route exists — the cases listed in each table.

General principle. Use the broadest scope: keep 1 wherever **any** production route is technically possible, even if it is uncommon. If in doubt, keep 1.

Mixtures (route = mixture)

Input → Output	DV	Why
020714 frozen chicken cuts → 160249 prepared pork and mixtures	stays 1	Another meat in a meat mixture
070310 onions → 200570 prepared olives	stays 1	Flavouring or stuffing
070960 capsicum → 200570 prepared olives	stays 1	Pimento-stuffed olives
551219 polyester woven fabric → 551693 rayon woven fabric	1→0	“Mixed” refers to fibre content; fabric → fabric of another fibre has no route

Minor constituents of textile goods (route = ingredient or mixture)

A garment, made-up article, yarn or fabric is classified by its main fibre, but it can contain other fibres: lining, pocketing, trim or blend. A fibre, yarn, fabric or waste of another fibre going into it stays 1:

- **certain 1** when the input is **cotton or a man-made fibre** (synthetic or artificial). These are the usual blend partners and lining materials, and the HS classifies a mixed good by the fibre that predominates

by weight, so a “cotton” garment can contain almost half man-made fibre;

- **1 with *uncertain* = 1** when the input is another fibre (wool, silk, flax, ramie, coarse animal hair, other textile materials), and when fabric of the **same fibre but another construction** goes into a garment (woven fabric → knitted garment and the reverse). These pairs count in DV but not in DV_strict.

Input → Output	DV	Why
540231 nylon filament yarn → 620342 men’s cotton trousers	stays 1	Man-made yarn as minor constituent
540752 dyed polyester woven fabric → 620311 men’s wool suits	stays 1	Lining or blend
520821 cotton fabric → 630253 table linen of man-made fibres	stays 1	Cotton as blend or trim
540231 nylon filament yarn → 520851 printed cotton fabric	stays 1	Minor yarn in a fabric
510710 wool yarn → 610463 women’s knitted synthetic trousers	1, uncertain	Wool blend: possible, less common
500720 silk woven fabric → 570110 wool carpets	1, uncertain	Silk as minor constituent
520811 cotton woven fabric → 610610 women’s knitted cotton blouses	1, uncertain	Same fibre, other construction (woven trim)

Recycling (route = recycle_waste; recycle_direct)

Input → Output	DV	Why
551431 synthetic woven fabric → 550510 synthetic fibre waste	stays 1	Cutting waste
411200 finished sheep leather → 411520 leather parings and dust	stays 1	Shaving and buffing waste
510310 wool noils → 510521 combed wool in fragments	stays 1	Waste reprocessed into fibre
721320 hot-rolled rod → 720610 non-alloy ingots	stays 1*	Direct remelting with no scrap code in between
761010 aluminium doors and windows → 760110 unwrought aluminium	stays 1*	Direct remelting
780500 lead pipes → 780110 refined lead	stays 1*	Direct remelting
540752 dyed polyester woven fabric → 540233 textured polyester yarn	stays 1*	Polyester can be melted and spun again

* Direct remelting is 1 in the main DV and coded *route* = recycle_direct. The robustness variable DV_excl_remelt sets these pairs to 0.

Broad scope: wastes, raw materials and uncommon uses

Input → Output	DV	Rule
292212 diethanolamine → 382520 sewage sludge	stays 1	Any chemical (ch. 28–38) → wastes of heading 3825 (recycle_waste)

290322 trichloroethylene → 382541 halogenated waste solvents	stays 1	As above
252400 asbestos → 252330 aluminous cement	stays 1	Any mineral of ch. 25 → cement (ingredient)
251200 diatomite → 252329 Portland cement	stays 1	As above
480451 kraft paper → 482320 filter paper	stays 1	Technically possible further use (forward)
910811 watch movements → 910599 other clocks	stays 1	Technically possible further use (component)

Feed (route = feed)

The input must be a feed-grade good — whole low-value fish (herring, sardines, mackerel, other fish), fish meat and meal, crustaceans and their meal (krill), molluscs — and the output a farmed carnivorous fish or crustacean (trout, salmon, eels, sea bass, flatfish, bluefin tuna, shrimps and prawns, crabs) or a product made from it. Wild-caught outputs (other tunas, skipjack, cod, herring, plaice) and grazers or filter feeders are 0. Where a feed use is possible but not established, the pair is 1 with *uncertain* = 1.

Input → Output	DV	Why
030264 fresh mackerel → 160411 prepared salmon	stays 1	Feed fish → farmed salmon
030371 frozen sardines → 030191 live trout	stays 1	Feed fish → farmed carnivore
030629 other crustaceans and crustacean meal → 030621 rock lobster	1, uncertain	Rock lobster is farmed on such feed in parts of Asia
030410 fresh fish fillets → 030613 frozen shrimp	1→0	Fillets are a human-food form, not feed
030350 frozen herring → 160414 canned tuna	1→0	Canned tuna is wild-caught
030710 oysters → 030332 frozen plaice	1→0	Oysters are high-value food, not feed
any fish → abalone, oysters or mussels	1→0	Grazers and filter feeders do not eat fish feed

Other routes

Input → Output	DV	Route
010639 live birds → 020890 other meat	stays 1	forward
120400 linseed → 200811 peanut butter	stays 1	ingredient
350211 dried egg albumin → 350219 liquid egg albumin	stays 1	reconstitute
290220 benzene → 290611 menthol (process solvent)	stays 1	ingredient (process)

Two AI coders and conflicts

- **Setup.** Two independent coders, ideally from different model families, receive the same codebook (steps 2–6 with these examples) and the same pair. Neither sees the other's answer.

- **Units.** To code tens of thousands of pairs, pairs are grouped by heading pair (input HS4 × output HS4). Each coder gives one default code per unit plus exceptions for single input codes, output codes or pairs, with the 5 most similar coded pairs as worked examples.
- **Question asked.** Each coder decides whether the pair should be turned from 1 to 0, and at which step, and returns {DV, route, step, one-sentence reason}.
- **Combining rule.** Both say 0: DV = 0. Both say 1: DV = 1. If they disagree, the pair stays 1 with *uncertain* = 1, and both codes and reasons are kept.
- **Decided by rule.** Where a scope convention (last section) settles a disagreement — for example intermediate siblings, chemicals → wastes, minerals → cement — the pair is coded by that convention with *uncertain* = 0.
- **Strict robustness DV.** Count only certain 1s: DV_strict = DV × (1 − uncertain), optionally restricted to forward, ingredient and component routes.
- **Reliability.** Report Cohen's kappa between the two coders. Then code a stratified sample of about 300 pairs across sources, steps and routes to estimate the accuracy of the final codes, weighting each stratum by its size. Without a domain expert, the sample is coded by two new AI sessions that see only the codebook and the pairs (no earlier answer), and a person reviews every pair where they disagree with the final code. Errors that form a group are corrected by rule in every PTA, not pair by pair. Report it as an independent AI audit, not as expert hand-coding.

Step 7. Scale up

(a) Exact matches

Any pair already in the coded master list (same HS6 pair after concordance) takes the existing DV, route and uncertain values, with source = exact_copy. Exact copies are re-copied whenever the master labels change, so a pair has the same code in every PTA.

(b) Train a model on the coded pairs

- Features: input and output chapters and headings; same-chapter and same-heading flags; flags from the step 2, 4 and 5 rules; text embeddings of both descriptions and their interaction.
- Target: whether the pair stays 1 or is turned to 0 (optionally also route and step).
- Validation: hold out whole chapter pairs (grouped cross-validation) so the model cannot memorise near-duplicate pairs; random cross-validation overstates accuracy.

(c) Apply with confidence bands

Predicted probability	Action
≥ 0.9	Keep at 1 (source = ml)
≤ 0.1	Turn to 0 (source = ml)
Between 0.1 and 0.9	Send through steps 2–6 with the two AI coders, giving them the 5 most similar coded pairs as worked examples

Support condition. Use the bands only where the chapter pair has more than 30 coded pairs in the master list, and only for pair types where the audit (d) shows the bands are reliable. Other pairs go to the AI coders or to explicit rules.

(d) Audit

Check a random sample of model-coded pairs from each band against the two AI coders (or a person). If accuracy in a band falls below the threshold, narrow that band and send more pairs to the AI coders. When the model and an explicit rule disagree, the rule is applied where it finds a route; otherwise the pair goes to the AI coders.

(e) Explicit rules for structured chapters

Textile chapters 50–63 are coded by explicit rules. Each code has a stage (fibre, yarn, fabric, trim, made-up or garment, waste), a fibre and, for fabrics and garments, a construction (knitted or not). The rules apply steps 2–6 and the conventions above: forward stage with the same fibre = 1; minor constituents as in step 6; finished siblings = 0; backward = 0; intermediate siblings of one fibre = 1; sewing thread and trims → articles = component; waste reprocessed = 1.

Reading “of other textile materials”. In the HS this means every fibre that is **not named** in the same group of subheadings. 610990 T-shirts “of other textile materials” include synthetic T-shirts, because only cotton has its own subheading (610910), so 540231 nylon yarn → 610990 is a main-fibre (forward) pair. 620449 dresses of other textile materials exclude wool, cotton, synthetic and artificial fibres, which have their own subheadings.

> **Example.** Another agreement restricts 030342 frozen yellowfin tuna → 160414 prepared tuna; the exact pair is already coded, so it is copied and stays 1 (forward). A new pair, 030389 other frozen fish → 160420 other prepared fish, has close neighbours coded 1 (forward, feed); the model gives p = 0.94 and it stays 1. A third pair, 230120 fishmeal → 020714 chicken cuts, falls in the middle band; the AI coders disagree (fishmeal as poultry feed), so it stays DV = 1 with uncertain = 1.

Summary of scope conventions (19 October 2026)

Question	Convention
Different intermediate goods of the same material at the same stage	1, route sibling_intermediate; finished siblings stay 0 (step 4)
Different synthetic fibres (nylon, polyester, acrylic, polypropylene)	One fibre
Fabrics of the same fibre, different construction	1 (sibling_intermediate)
Sewing thread ↔ yarn of the same fibre	1 (sibling_intermediate; thread → twine forward)
Clothing accessories (trims, bows) → garments	1 (component)
Any chemical (ch. 28–38) → wastes of heading 3825	1 (recycle_waste)
Any mineral of ch. 25 → cement (2523)	1 (ingredient)
Technically possible but uncommon further use	1
Minor constituent of another fibre in a garment, made-up, yarn or fabric	Cotton or man-made input: 1. Other fibres, or the same fibre in another construction (fabric → garment): 1 with uncertain = 1
Direct remelting with no scrap code in between	1, route recycle_direct; 0 in DV_excl_remelt
Fish and other seafood of different species	0 (step 3), unless the input could feed the output: feed (1); possible but unestablished feed: 1 with uncertain = 1
“Of other textile materials”	Every fibre not named in the same group of subheadings
The two AI coders disagree and no convention settles it	1 with uncertain = 1